

NPTEL

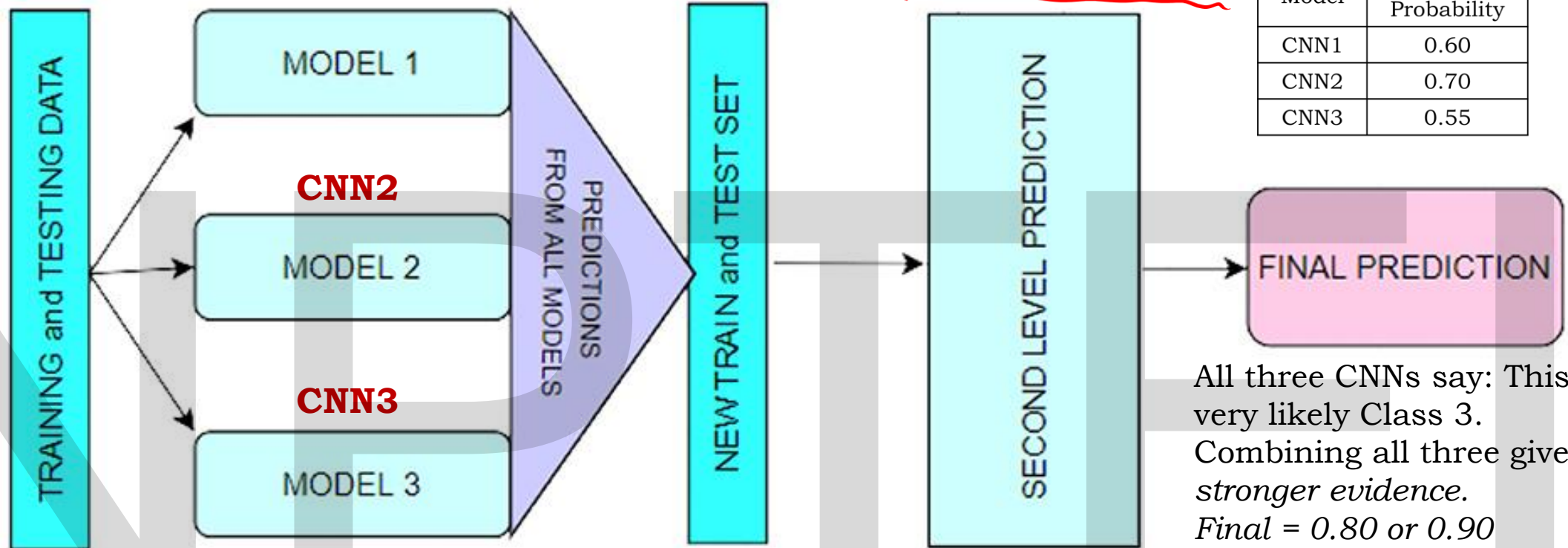
Ensemble Deep CNN

Alzheimer dataset

Stacking_Model = Model(inputs=stack_input, outputs=stack_output)

CNN1 [0.10, 0.20, 0.60, 0.10] **MLP/Any model**

Model	Class 2 Probability
CNN1	0.60
CNN2	0.70
CNN3	0.55



All three CNNs say: This image is very likely Class 3.
Combining all three gives *stronger evidence*.
Final = 0.80 or 0.90

FIRST LEVEL MODEL [0.15, 0.10, 0.70, 0.05] merged → [0.10, 0.20, 0.60, 0.10, 0.15, 0.10, 0.70, 0.05, 0.05, 0.25, 0.65, 0.05]
[0.05, 0.25, 0.65, 0.05]
merged = Concatenate()([pred1, pred2, pred3])

	CNN 1				CNN 2				CNN 3				
Image	CNN1_P1	CNN1_P2	CNN1_P3	CNN1_P4	CNN2_P1	CNN2_P2	CNN2_P3	CNN2_P4	CNN3_P1	CNN3_P2	CNN3_P3	CNN3_P4	True Label
Img1	0.10	0.20	0.60	0.10	0.15	0.10	0.70	0.05	0.05	0.25	0.65	0.05	AD
Img2	0.80	0.10	0.05	0.05	0.75	0.12	0.10	0.03	0.82	0.08	0.07	0.03	CN
Img3	0.12	0.20	0.55	0.13	0.10	0.15	0.65	0.10	0.08	0.18	0.62	0.12	MCI
Img4	0.05	0.10	0.10	0.75	0.08	0.12	0.15	0.65	0.10	0.11	0.18	0.61	Moderate

	CNN 1				CNN 2				CNN 3				
Image	CNN1_P1 ✓	CNN1_P2	CNN1_P3	CNN1_P4	CNN2_P1	CNN2_P2	CNN2_P3	CNN2_P4	CNN3_P1	CNN3_P2	CNN3_P3	CNN3_P4	True Label
Img1	<u>0.10</u>	0.20	0.60 ✓	0.10	<u>0.15</u>	0.10	0.70	0.05	<u>0.05</u>	0.25	0.65	0.05	AD
Img2	0.80	0.10	0.05	0.05	0.75	0.12	0.10	0.03	0.82	0.08	0.07	0.03	CN
Img3	0.12	0.20	0.55	0.13	0.10	0.15	0.65	0.10	0.08	0.18	0.62	0.12	MCI
Img4	0.05	0.10	0.10	0.75	0.08	0.12	0.15	0.65	0.10	0.11	0.18	0.61	Moderate

CNN1_P1=0.10 ✓

CNN2_P1=0.15 ✓

CNN3_P1=0.05 ✓

All these are low. The MLP learns low probabilities across all. Do not predict class1. This reduces false positives.

CNN1_P3=0.60 ✓

CNN2_P3=0.70 ✓

CNN3_P3=0.65 ✓

All three models agree. The MLP learns very strong evidence for class3.

The stacking meta-learner learns:

- **How much to trust each model,** ✓
 - **for which class,** ✓
 - **in which situation,** ✓
 - **and in what combination —** ✓
- to produce the best final prediction.** ✓

This is the essence of stacking.